B.TECH 2025-26 COURSE SYLLABUS

Semester-II

Course Code: CS1807 **Name of the Programme: B.Tech (H)** **Name of the Course: Linear Algebra** **Semester: 2**		
Course Credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
3	2-1-0	45
Course Objectives:		
A. Introduce solutions to systems of linear equations using Gauss Elimination, Gauss-Jordan Elimination and Gauss-Seidel method.		
B. Describe fundamental concepts to matrix subspaces like span and linear independence and rank-nullity theorem.		
C. Illustrate linear transformations and describe and solve eigenvalue problems.		
D. Describe orthogonal basis sets and construction of one using Gram-Schimdt process to set the stage for QR decomposition, principal axis theorem, SVD and spectral theory.		

Syllabus:

Module – 1: Linear Equations	**No. of hours**
	6
System of linear equations; row echelon form; consistent and inconsistent equations; Gaussian elimination; reduced row echelon form; Gauss-Jordan Elimination, Inverse; LU decomposition; Gauss-Seidel method.	

Module – 2: Span and subspaces	**No. of hours**
	6
Homogeneous equations; span; linear independence/dependence of basis; subspaces; dimension, rank and nullity; rank-nullity theorem.	

Module – 3: Linear transformations	**No. of hours**
	6
Linear transformations, domains; codomains; image; range and kernel of a linear map; composition of linear maps; Inverse of a linear transformation; eigenvalue equation; Characteristic equation; eigenvalue and eigenvectors; eigenspaces; similarity and diagonalization.	

Module – 4: Orthogonality	No. of hours 6
Orthogonality; orthogonal and orthonormal: -basis, matrices, projections, decomposition; Gram-Schmidt orthogonalization; QR decomposition; orthogonal diagonalization;	

Module – 5: Spectral Theorem	No. of hours 6
Spectral theory; quadratic forms; principal axis theorem; norm and distance; least squares approximation; Singular Value Decomposition.	

Course Outcomes
a. Apply Gaussian elimination, LU decomposition and others to solve systems of linear equations.
b. Compute the span, basis and dimension of matrix subspaces to solve related problems in computer science.
c. Compute eigenvalues and eigenvectors of a given matrix to solve real world problems.
d. Apply orthogonal projections, Gram-Schmidt processes, and Singular Value Decomposition to solve approximation problems.

Text Books	
1	D. Poole, *Linear Algebra: A Modern Introduction*, 4th ed. Boston, MA, USA: Cengage Learnin 2014, ISBN: 978-8131530245.
2	G. Strang, *Introduction to Linear Algebra*, 5th ed. Wellesley, MA, USA: Wellesley-Cambridge Press, 2021, ISBN: 978-1733146654.

Reference Books	
1	S. Lipschutz and M. Lipson, *Schaum's Outline of Linear Algebra*, 6th ed. New York, NY, USA: McGraw-Hill Education, 2018, ISBN: 978-1260011456.
2	S. Boyd and L. Vandenberghe, *Introduction to Applied Linear Algebra*, Cambridge, U.K.: Cambridge Univ. Press, 2018. [Online]. Available: https://web.stanford.edu/~boyd/vmls/vmls.pdf. ISBN: 978-1-316-51896-0.
3	D. C. Lay, S. R. Lay, and J. J. McDonald, *Linear Algebra and its Applications*, 5th ed. Boston, MA, USA: Pearson, 2016, ISBN: 978-0-321-98238-4.
4	F. Neri, *Linear Algebra for Computational Sciences and Engineering*, Cham, Switzerland: Springer, 2019, ISBN: 978-3-030-21321-3.
5	E. Kreyszig, *Advanced Engineering Mathematics*, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2006, ISBN: 9780471728979.